

# EEG Pipeline – Setup Guide

Emotiv EPOC X (14 channels) | MATLAB + EEGLAB | runs entirely offline

## What this does

Takes raw EEG recordings, automatically removes noise and artifacts (drift, blinks, muscle, bad electrodes), saves the cleaned data, and produces power spectrum plots from 0–20 Hz for all 14 channels. Nothing is ever uploaded — all processing happens on this computer.

## Step 1 • Install MATLAB

MATLAB R2021a or newer, with these two toolboxes:

- Signal Processing Toolbox
- Statistics and Machine Learning Toolbox

## Step 2 • Install EEGLAB

Download from [eeglab.org](http://eeglab.org) and unzip it into the **toolboxes** folder inside this project. Then add these four plugins from inside EEGLAB, via *File > Manage EEGLAB extensions*:

- **BIOSIG** — reads EDF/BDF files
- **firfilt** — the filters
- **clean\_rawdata** — removes large artifacts (ASR)
- **ICLabel** — identifies blinks and muscle noise

## Step 3 • Add your recordings

Each dataset gets its own subfolder inside **data\raw**, e.g. `data\raw\Harvard\...` Accepted formats:

- **.edf** or **.bdf** — preferred
- **.csv** — EmotivPRO export
- **.dat** — headerless matrix; sample rate and channel order must be given explicitly, since the file carries neither

Downloaded a zip? `import_dataset_zip('path\to\file.zip')` extracts it into its own correctly-named subfolder in one step.

This folder is never modified. Original recordings stay untouched.

## Step 4 • Start it

Double-click **1 - START HERE.bat**, or open MATLAB and run:

```
cd('<this folder>')
addpath('src')
cfg = setup_paths();
check_env();
```

`check_env()` lists anything still missing. If it says “All hard requirements met”, you are ready.

## Step 5 • Process everything

```
results = run_pipeline();
```

That is the whole thing. It processes whichever dataset subfolder was extracted most recently, cleans every recording, saves the cleaned data, plots each spectrum, and builds the three-recording comparison figure. Pass 'Dataset', 'name' to pick a specific one instead.

## Where the results appear

|                               |                                                                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>figures\</b>               | Power spectrum plots, plus the three-recording comparison                                                                       |
| <b>results\qc_summary.csv</b> | Open this first. One row per recording, listing every decision made. The <b>usable</b> column: 1 = fine, 0 = too noisy to trust |
| <b>data\processed\</b>        | Cleaned recordings, one .set file each                                                                                          |

**Adjusting the pipeline.** Every filter setting, threshold and on/off switch lives in `src\pipeline_config.m`, each with a comment explaining why it has the value it does. Cleaning thresholds are **adaptive by default** — computed per recording rather than fixed — so they travel to a new dataset without retuning. Nothing needs editing anywhere else.

**Note on short recordings.** ICA and ASR need roughly 30–45 seconds of data to work reliably. On shorter recordings the pipeline skips them and says so, rather than producing a result that cannot be trusted.